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HO CHI MINH CITY CAMPUS

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Manga Creation Workflows & Publishing Management System
Software Requirement Specification

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Table of Content

1. Project Context.....	3
1.1 Manga Industry Background.....	3
1.2 Problems.....	3
1.3 Systems Objectives.....	4
1.4 Core Platform Nature.....	4
1.5 Access Boundaries & User Group.....	4
1.6 Definitions, Acronyms, and Abbreviations.....	5
2. Scope.....	5
2.1 In-Scope.....	5
2.2 Out of Scope.....	6
3. Context Diagram.....	7
4. Actors & Features.....	7
4.1 Mangaka.....	7
4.2 Assistant.....	10
4.3 Tantou Edit.....	11
4.4 Editorial Board.....	12
4.5 System Admin.....	13
5 Main Flows.....	14
5.1 Main Flow 1: Create Mangaka Account.....	14
5.2 Main Flow 2: Register & Approval Assistant Account.....	15
5.3 Main Flow 3: Submit & Review a New Series.....	17
5.4 Main Flow 4: Task Assignment and Lock/Transfer Payment Processing.....	17
5.5 Main Flow 5: Edit Chapter & Disburse Royalties.....	19
5.6 Main Flow 6: Rating & Maintaining Publication.....	20
5.7. Main Flow 7: Top-up, Withdrawal, and View Wallet Dashboard.....	21
6. Business Rules.....	23
6.1. Account & Permission Rules.....	23
6.2. Review & Funding Rules.....	23
6.3. Task Management & Sub-Contract Rules.....	24
6.4. Review Genkōryō Rules.....	25
7. Technologies.....	26
Frontend.....	26
Backend.....	27
Database & Storage.....	28
AI Service.....	29
DevOps & Tooling.....	29
8. Data Model.....	30
8.1. Conceptual ERD.....	30
8.2. Logical ERD.....	31
9. User Guide.....	32
9.1. System Overview.....	32
9.2. System Specification.....	32
Flow 1 — Onboarding & Account Creation.....	32
Flow 2 — Series Review & Production Funding.....	33
Flow 3 — Production & Hiring Assistants.....	34
Flow 4 — Editing, Genkouryou & Publishing.....	35
Flow 5 — Ranking & Publication Maintenance.....	35
9.3. Wallet & Payment Management.....	36

1. Project Context

1.1 Manga Industry Background

The Japanese and global manga industries operate on a highly structured, collaborative production pipeline. Getting a manga from an artist's drawing board to final publication requires a strict sequential workflow involving several key stakeholders:

Within traditional publishing houses (such as Shueisha, Kodansha, etc.), this process follows these stages:

- **Creation & Delegation:** The lead Mangaka drafts the initial layouts (storyboards) and delegates tedious sub-tasks (drawing backgrounds, applying screen tones, adding special effects, basic color-filling) to a team of assistants.
- **Editorial Audit:** The assigned Tantou Editor supervises the Mangaka, reviewing page drafts for plot consistency, pacing, and visual quality.
- **Evaluation & Approval:** New series proposals must be formally reviewed and voted upon by the Editorial Board before getting approved for serialization.
- **Publication & Feedback:** Approved chapters are published in physical or digital magazines (Weekly/Monthly). Reader feedback and voting cards are compiled to track rankings.
- **Continuation Decision:** Series that consistently place low on reader ranking lists over several weeks face cancellation (axing).

1.2 Problems

Currently, manga production workflows rely heavily on legacy or fragmented communication tools, causing several operational inefficiencies:

- **Fragmented Collaboration Tools:** Mangakas and Assistants coordinate using unrelated platforms (such as Email, LINE chat, Dropbox, and Google Drive) resulting in lost assets and severe communication gaps. Affected Stakeholders: Mangaka, Assistant.
- **Lack of Progress Control:** Lead artists cannot easily monitor what percentage of a page an assistant has completed or identify which specific drawing pages are currently facing delays. Affected Stakeholders: Mangaka, Assistant.
- **Absence of Visual Review Tools:** Tantou Editors must print pages physically or write textual descriptions on flat PDFs to note mistakes, making it difficult to convey precise correction coordinates. Affected Stakeholders: Tantou Editor.

- Manual Evaluation Workflows: The Editorial Board conducts offline meetings and uses paper ballots to vote on proposals. There is no centralized digital registry to log historical decision records. Affected Stakeholders: Editorial Board.
- Fragmented Reader Ranking Data: Readers' physical vote cards are input manually by staff, leading to long data aggregation times and delayed decision-making. Affected Stakeholders: Editorial Board.

Consequence: Frame partitioning (segmentation) and basic background color filling consume a substantial amount of an artist's creative time. The entire production chain lacks a centralized "Single Source of Truth" to track real-time progress and automate financial transactions.

1.3 Systems Objectives

- Centralized Digital Workspace: Build a unified web application that digitizes the entire manga pipeline from initial drafting, task delegation, and editorial reviews to voting and final publication.
- Deep Learning Automation (AI): Deploy artificial intelligence services to automate page-panel segmentation and assist in colorization, speeding up drawing times and optimizing artists' efforts.

1.4 Core Platform Nature

- B2B Workspace: MCWPMS is NOT a public, mass-market manga reader website (B2C).
- Digital Publishing Workspace: It is a B2B Collaborative Workspace & Publishing Management Platform operating as the digital workspace for a publishing house. It coordinates remotely contracted Mangakas and enables those creators to safely hire and manage freelance Assistants.

1.5 Access Boundaries & User Group

The system strictly separates users into two distinct operational environments:

- A. Internal Group (Accounts created strictly by Admins)
 1. System Admin: Manages system parameters, assigns editors to Mangakas, configures core master catalogs, and creates system credentials.
 2. Tantou Editor: Direct employees of the publisher. Oversees specific authors, reviews manuscripts, tracks progress, and sets chapter deadlines.
 3. Editorial Board: High-level executives of the publisher who vote on series

approvals and monitor overall performance rankings.

A. External Group (Interact via the platform)

1. Mangaka: Lead artists signed under contract with the publisher. Accounts are created and activated by System Admins.
2. Assistant: Freelance graphic artists. This is the only role allowed to self-register (Signup) on the platform's landing page to apply for active drawing tasks and receive wages.

1.6 Definitions, Acronyms, and Abbreviations

Acronym	Definition
MCWPMS	Manga Creation Workflows & Publishing Management System
SAM	Segment Anything Model (used for smart canvas region selection)
Z-Index	Layers depth coordinate determining graphic placement order on canvas
Genkōuryō	Standard page-rate fee paid to authors for manuscript submissions.
RBAC	Role-Based Access Control

2. Scope

2.1 In-Scope

- **Account Management:** Profile setup, dashboard tracking, and Role-Based Access Control (RBAC) separating Internal and External roles.
- **Series & Manuscript Management:** Uploading and tracking draft chapters and manuscript packages.
- **Task & Assistant Management:** Canvas-based panel region selection, task division, and assistant verification.
- **Editing & Review:** Interactive visual canvas annotation, progress tracking, and deadline controls.
- **Editorial Board & Publishing:** Series defense voting, automated rankings, and publication calendar.
- **Wallet & Internal Payment:** Transactional Lock / Transfer / Unlock flows, Genkōuryō payouts, and VNPay Sandbox banking simulator.
- **AI Integration:** Deep learning microservice for auto-segmenting panels,

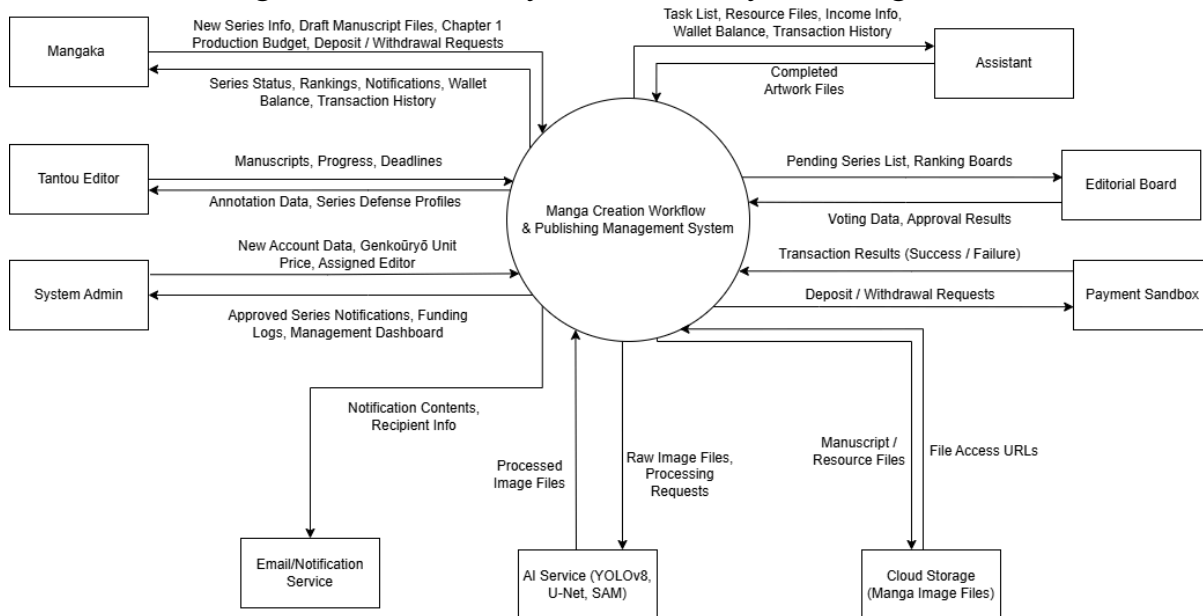
auto-compositing layer submissions based on Z-indices, and colorization assistance.

2.2 Out of Scope

- B2C Reader Distribution Platform (Nền tảng phân phối độc giả): The platform is strictly an internal back-office collaborative workspace. It does not provide a public mass-market B2C web reader or mobile reading application for end-consumers.
- Native Mobile Applications (Phát triển Ứng dụng di động): The system is built as a responsive web platform accessible via mobile browsers. Developing independent native iOS or Android mobile applications is out of scope.
- Royalty Revenue Sharing (Tiền bản quyền theo % doanh thu): Tracking and calculating book sales or digital subscription royalty percentages is handled by external retail marketplaces and is not calculated inside this system's wallet logic.
- Production Payment Gateway (Cổng thanh toán thương mại): Real financial transaction processing over commercial credit cards or real bank accounts is out of scope. The internal wallet relies on a simulated VNPAY Sandbox API strictly for demonstration (demo) of deposit and withdrawal workflows.
- Offline Drawing Tools (Công cụ vẽ offline): The system does not provide interactive painting, sketching, or canvas creation tools. Artists draw offline on local applications (e.g., Photoshop, Clip Studio Paint) and upload their raster graphic results.

3. Context Diagram

The Context Diagram defines the system boundary, showing the external actors that inter



the system and the major data flows between them.

act with

4. Actors & Features

4.1 Mangaka

Mangaka				
ID	Feature Name	Module	Description	Acceptance Criteria
F1.1	Create Series Profiles	Series	Enter name, genre, synopsis, upload sample artwork, and input "Chapter 1 Production Budget" (VND).	The series is successfully saved with status set to Draft.
F1.2	Submit Draft Manuscript	Series	Upload draft page images/PDF of the sample chapter.	Manuscript files are securely saved on Cloud Storage.

Mangaka

F1.3	Submit for Review	Series	Transition the Series status from Draft to Pending_Approval.	The assigned Editor receives a system Notification.
F1.4	View Series Status	Series	Track the review progress: Pending / Approved / Rejected.	The system displays the correct status updates in Realtime.
F2.1	Upload Manga Pages	Task Management	Upload individual drawing pages of a new chapter.	Page files render smoothly on the Canvas Viewer.
F2.2	Canvas Region Selection	Task Management	Use the Region Selection Tool to highlight specific panel/drawing coordinates on a page.	Selected region coordinates are successfully saved in JSON format.
F2.3	Assign Tasks to Assistants	Task Management	Delegate the selected page region to an Assistant, configuring the pay rate, Deadline, and canvas layer level (Z-Index).	The system automatically locks the task fee in escrow (Lock state) and notifies the Assistant.
F2.4	View Composited Canvas	Task Management	Preview the compiled manga page after all assigned Assistants have submitted their completed artwork layers.	The merged/composited image (Composite Image) renders with correct layering and alignment.
F2.5	Approve/Reject	Task Management	Click Approve to accept the assistant's work, or click Reject and attach a visual Canvas Annotation describing required corrections.	Task status updates; locked funds are automatically transferred (Transfer state) to the Assistant's wallet upon approval.
F1.5	Track Series Ranking	Ranking	View the series rank position on the periodic ranking board.	The statistical charts or tables display correct, up-to-date data.

Mangaka

F1.6	Receive Alert Notifications	Notification	Receive warning alerts when the series ranking falls low, indicating a risk of series cancellation (axing).	The system delivers push notifications in Realtime.
F1.7	View Wallet Dashboard	Wallet	Monitor balances across two wallet allocations (SetupFundBalance & WithdrawableBalance) and view the transaction history.	Dashboard hiển thị chính xác, realtime. The financial dashboard displays accurate, real-time ledger data.
F1.8	Withdraw Funds	Wallet	Request a payout withdrawal from the withdrawable wallet balance to a bank account using the simulated VNPay Sandbox.	Upon a successful simulated transaction, the withdrawable wallet balance updates instantly.
F1.9	Deposit Funds	Wallet	Deposit personal funds into the wallet balance using the VNPay Sandbox gateway when additional capital is needed to hire assistants.	Upon a successful sandbox payment, the wallet balance is credited and updated instantly.
F1.10	Approve Deadline Extension	Task Management	Approve or Reject a task deadline extension request submitted by an Assistant.	If approved, the system updates: New Deadline = Original Deadline + Requested Days.
F2.13	Manual Composite Tuning	Task Management	Manual Composite Edit: The Mangaka can drag and drop layers, rearranging the display order (Z-Index) on the Canvas before saving the final manuscript page.	The UI displays clear visual layer separation and performs adjustments without lag.
F2.14	Set Absence Status	Task Management	Toggle the user state to On_Leave (Absence/Emergency leave mode).	The system automatically freezes the Auto-Approve countdown timer for

Mangaka

active tasks to protect the author's escrow funds.

4.2 Assistant

Assistant

ID	Feature Name	Module	Description	Acceptance Criteria
F2.6	View Task Queue	Task Management	View assigned tasks. Filter by Status, and sort by Deadline/Priority.	Displays the user's correct workflow queue accurately.
F2.7	Download Files & Resources	Task Management	Download the original manuscript page (BaseLayer) and the resource folder (containing brushes, style guides).	Files download intact without corruption or incorrect compression formats
F2.8	Upload Drawing Submission	Task Management	Upload the completed drawing file for the assigned region.	The uploaded file must be in transparent PNG format; a notification is automatically sent to the Mangaka.
F2.9	Track Completed Tasks	Income	View the list and total number of Approved tasks grouped monthly.	The reporting dashboard displays accurate, correct counts.
F2.10	Track Income Statistics	Income	Monitor total earnings (sum of Payment_Amount) calculated monthly.	The income statistics chart displays data visually and accurately.
F2.11	Manage Wallet	Wallet	View wallet balance, transaction payment history, and request	The displayed balance matches the transaction ledger table, and

4.3 Tantou Edit

Tantou Editor				
ID	Feature Name	Module	Description	Acceptance Criteria
F3.1	Manga Viewer	Review	View the fully completed chapter, flip pages similar to a real reading experience.	Render is smooth and supports panning and zooming (Pan/Zoom).
F3.2	Annotation QC Tool	Review	Mark error regions on pages. Errors must be classified according to the standard QC taxonomy: Technical, Art, Content.	Displays error coordinates and error type. If errors exist, the page is flagged as Invalid.
F3.3	Manage Series Profile	Series	Compile and aggregate series metrics to defend the project before the Editorial Board.	Generate an evaluation report containing Ranking and user Engagement metrics.
F3.4	Progress Dashboard	Review	Monitor production progress, detailing count of Done, In-progress, and Pending pages.	The progress reporting dashboard updates states in Realtime.

Tantou Editor				
F3.5	Deadline Tracking	Review	Monitor print-submission deadlines and warn Mangakas who are lagging.	Trigger warning notifications (Alert) when the remaining time is 2 days or less.
F3.6	Approve Chapter	Review	Validate items on the QC Checklist before finalizing the count of approved pages (PageValidCount).	Clicking approve transitions the chapter status to Approved and automatically triggers the Genkōuryō (manuscript fee) payout.
F3.7	Dispute Resolution	Task Management	Access file version history and comments to decide resolution outputs (disbursing 100% to Assistant, refunding 100% to Mangaka, or making a partial payment split).	Escrow funds are automatically unlocked or transferred (Unlock/Transfer) by the system according to the split ratio finalized by the Editor.

4.4 Editorial Board

Editorial Board				
ID	Feature Name	Module	Description	Acceptance Criteria
F4.1	View Pending Series	Publishing	Access the list of series in Pending_Approval state along with evaluation profiles submitted by Editors.	Render list view integrates filter and search functionalities.
F4.2	Cast Vote	Publishing	Vote Approve/Reject, input a proposed production budget amount, and add qualitative comments.	Each board member is strictly restricted to voting exactly once per series.

Editorial Board

F4.3	Release Schedule Decision	Publishing	Assign a Weekly or Monthly publication schedule for approved series.	The publication schedule is permanently written into the database.
F4.4	Input Vote Data	Ranking	Manually input or import reader voting card data collected from external channels.	Enforces strict data validation checks, and records save successfully.
F4.5	View Ranking Board	Ranking	View the aggregated ranking dashboard updated after each voting data cycle.	Ranking charts refresh immediately upon data updates.
F4.6	Cancel Series	Publishing	Cast a vote to cancel serialization (Hủy) for series that consistently maintain low ranking statistics.	The series status updates to Cancelled; triggers clean-up routines for active tasks and sends notifications.
F4.7	Modify Publication Status	Publishing	Change the release schedule frequency or publication format (Weekly ⇔ Monthly).	Updates take effect instantly across the system.

4.5 System Admin

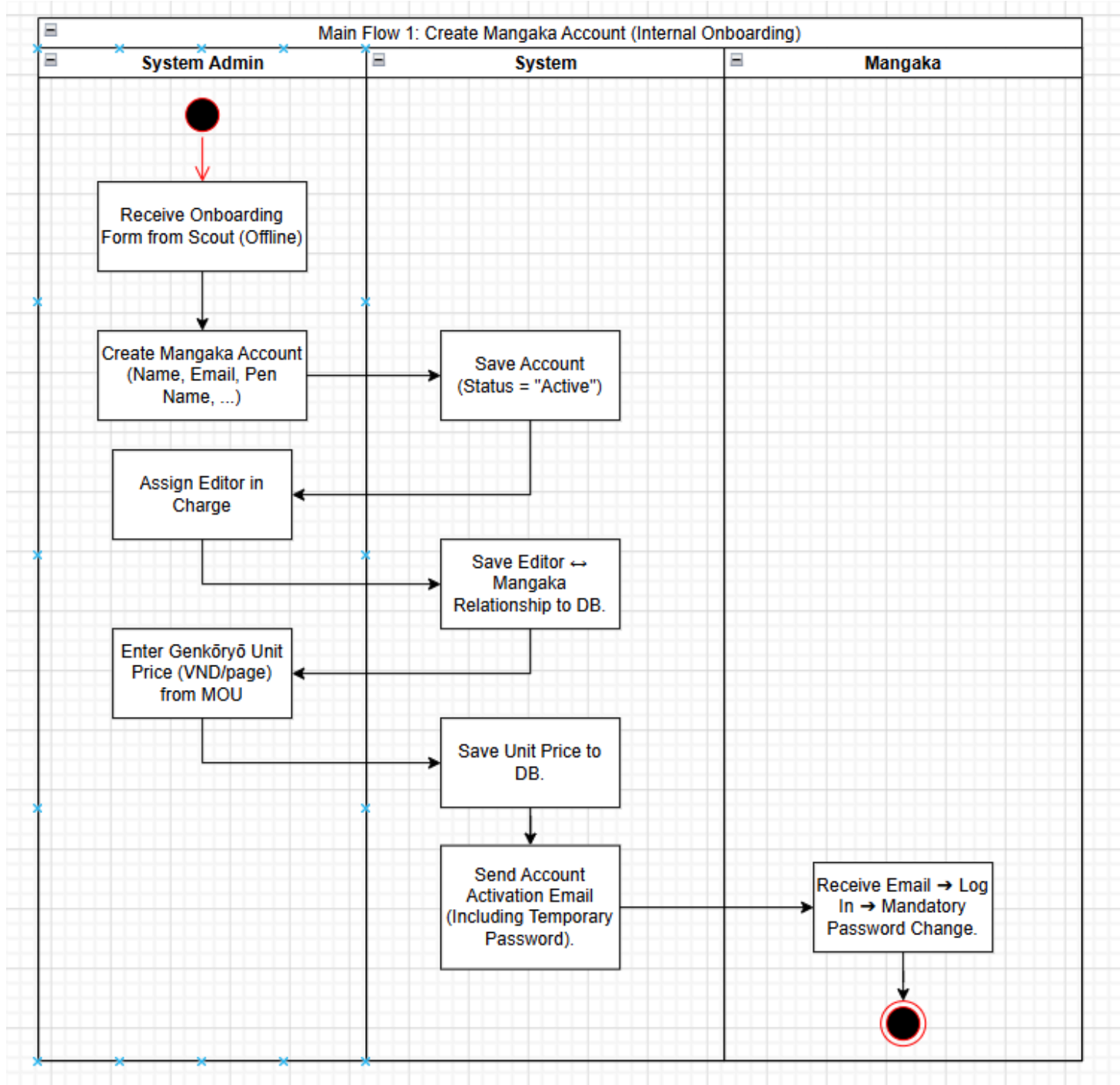
System Admin

ID	Feature Name	Module	Description	Acceptance Criteria
F5.1	Approve Accounts	User Management	Review the portfolios uploaded by Assistants and transition their account status to Active.	Users are only permitted to log in after the Admin has updated their status to Active.

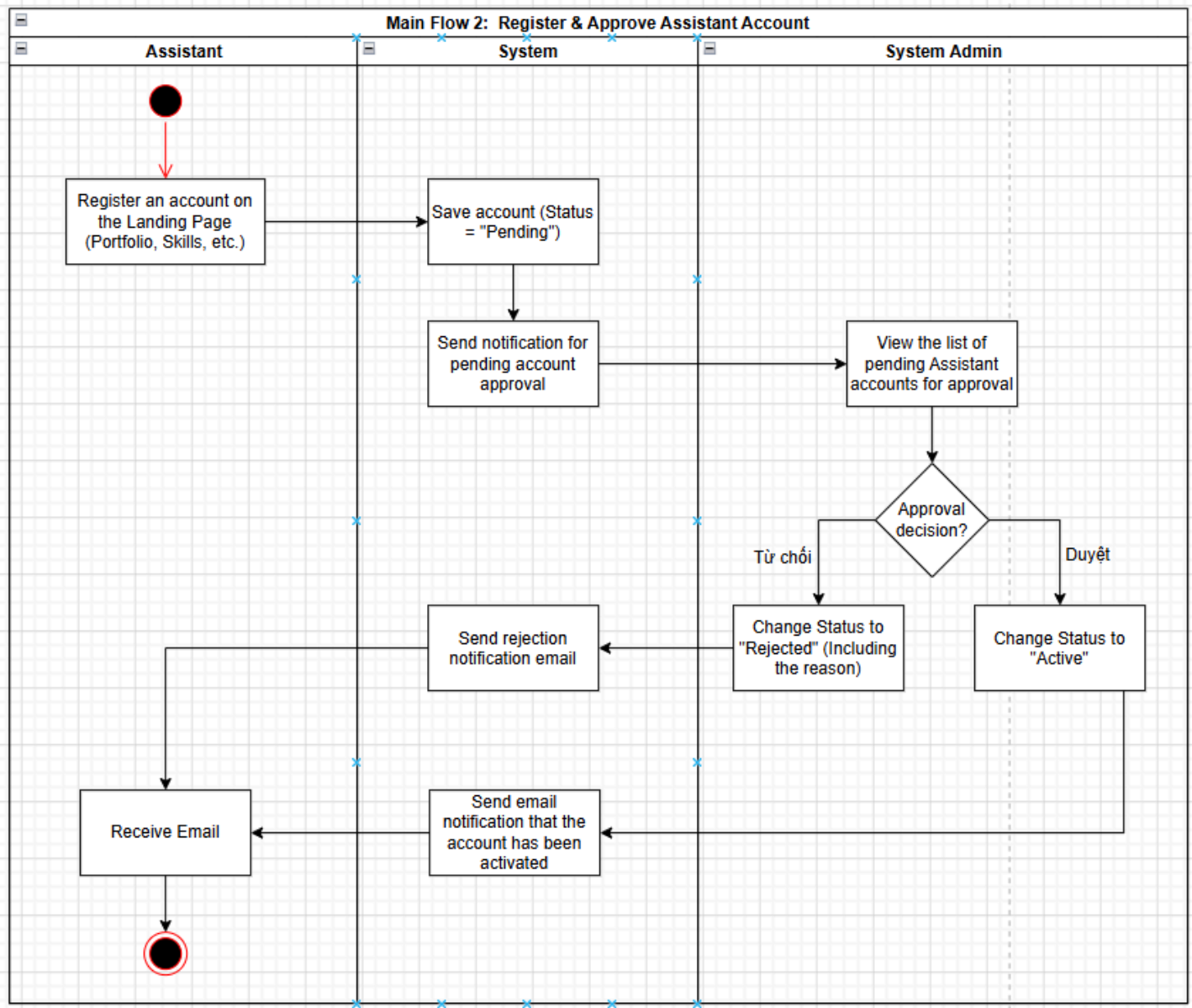
System Admin				
F5.2	Manage Permissions (RBAC)	User Management	Set role boundaries and access levels (RBAC) for user groups: Mangaka, Assistant, Editor, Editorial Board.	Users can only execute API features mapped to their JWT role claims.
F5.3	Input Base Page Rate	User Management	Save the baseline manuscript page fee (BaseGenkouryoPrice) specified in the Memorandum of Understanding (MOU) to the DB during contract setup.	The baseline price is saved and referenced for all subsequent automated payout calculations.
F5.4	Disburse Production Budget	Wallet	The system automatically credits the startup budget into the Mangaka's wallet immediately after the Mangaka clicks Accept Fund following a Board decision. Admin can view transaction logs.	The system transaction log comprehensively records the Funding operation.
F5.5	Update Contract Addendum	User Management	Record a revised page rate in the contract amendment log (ContractAddendum).	The newly updated rate is only applied to chapters created after the modification timestamp (non-retroactive).
F5.6	VNPay Reconciliation	Wallet	Reconciliation: Cross-reference aggregate deposit and withdrawal transaction records on the platform with settlement CSV spreadsheets exported from VNPay.	Flags and highlights transactions with mismatched transaction statuses for manual review.

5 Main Flows

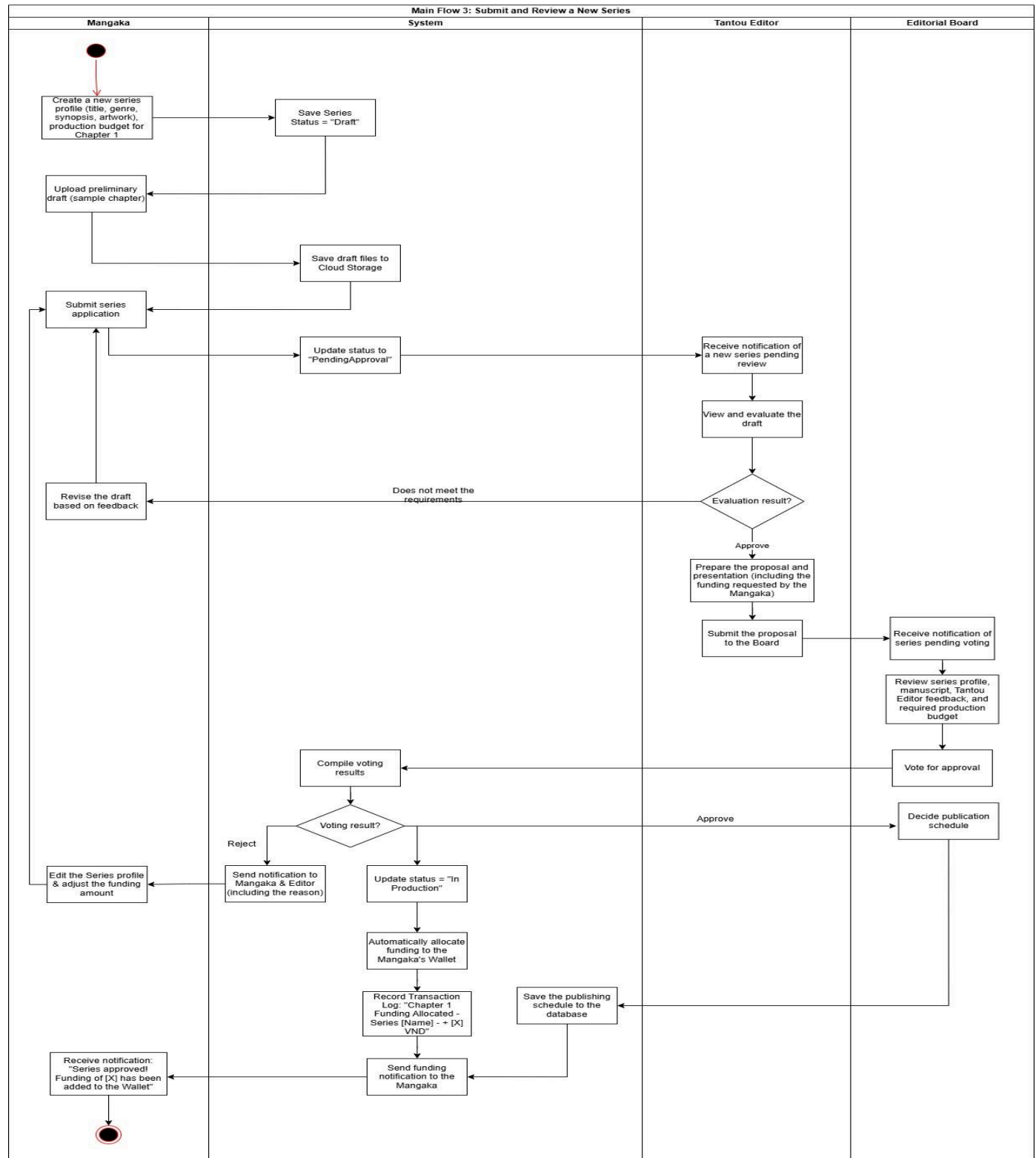
5.1 Main Flow 1: Create Mangaka Account



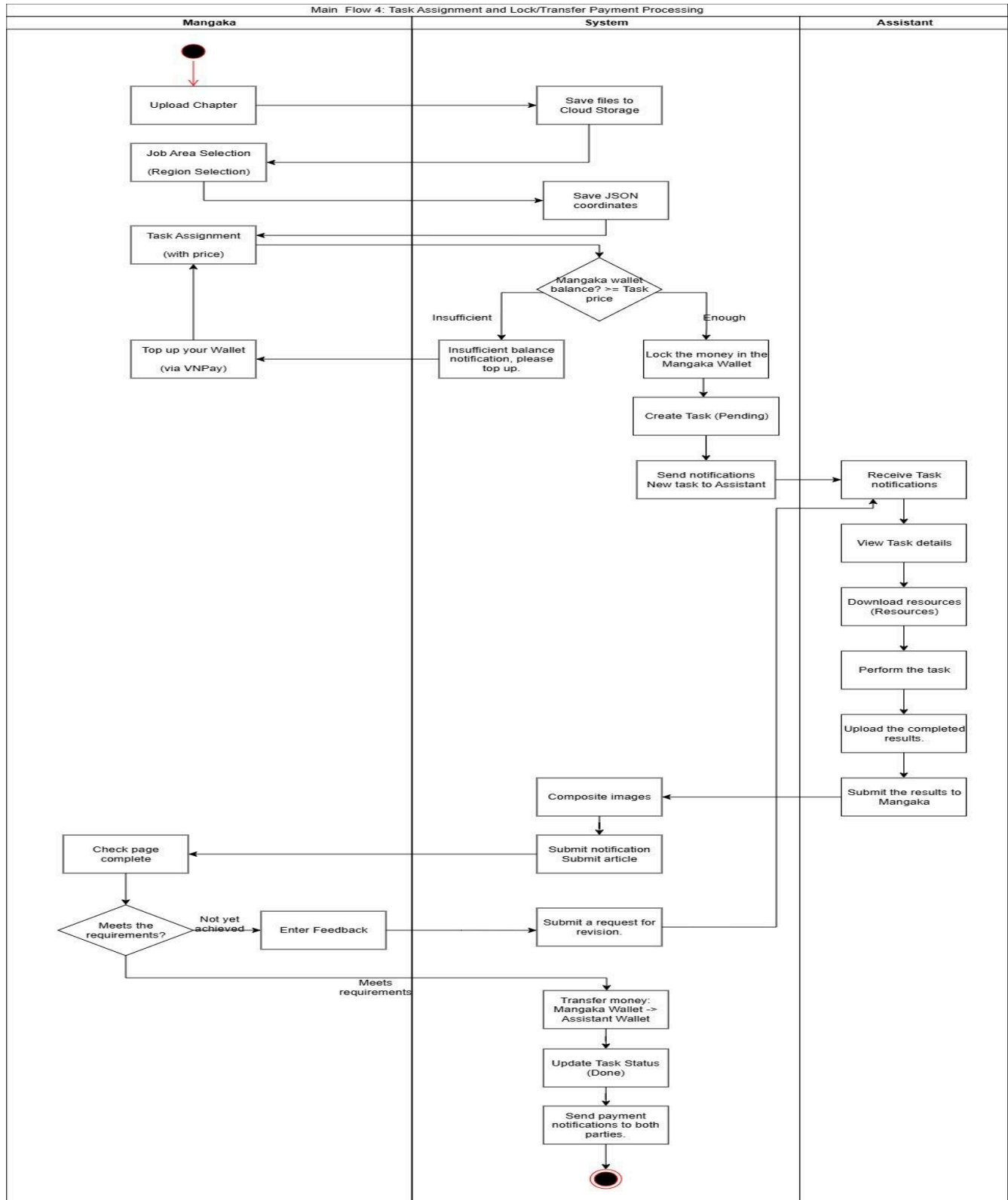
5.2 Main Flow 2: Register & Approval Assistant Account



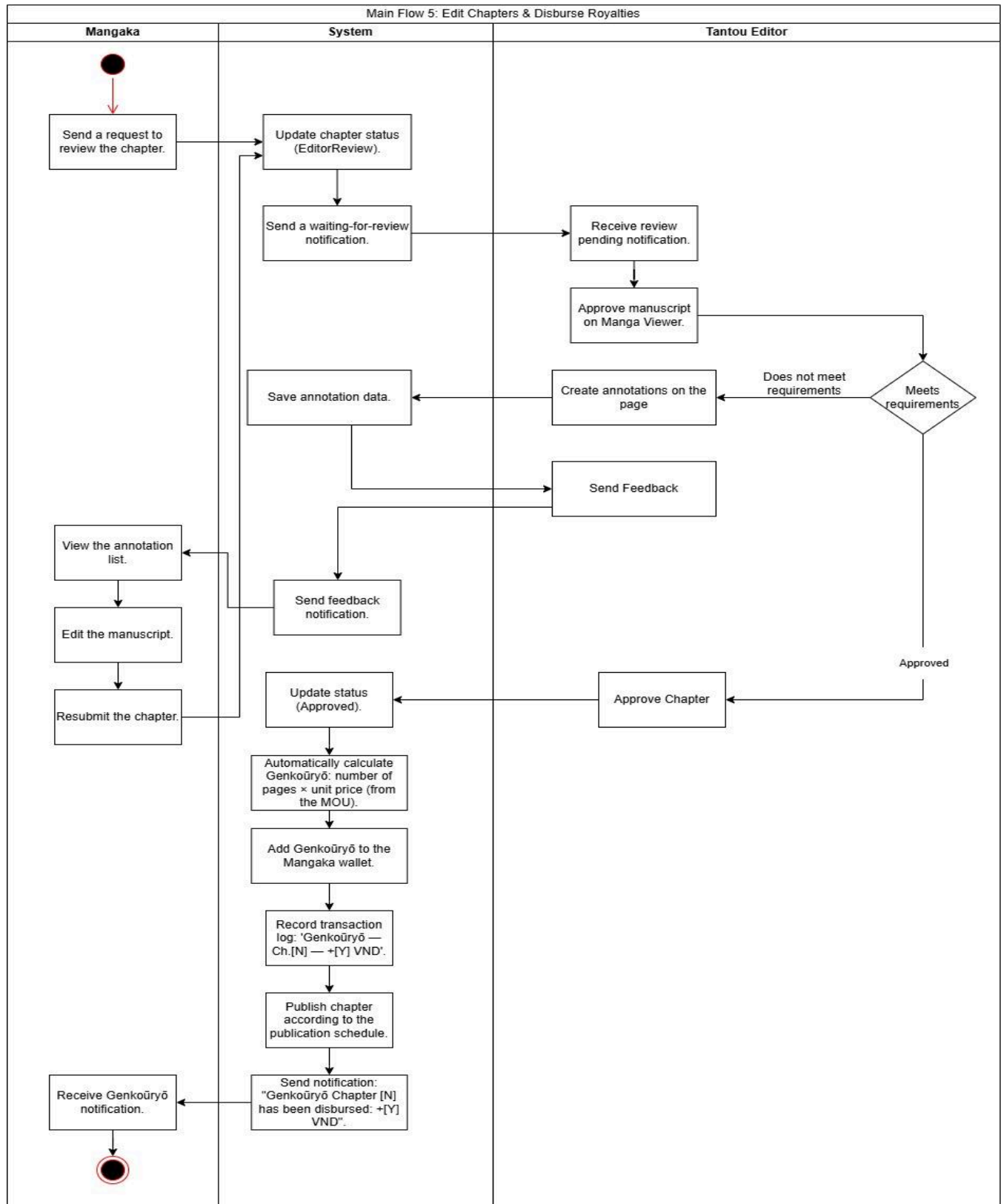
5.3 Main Flow 3: Submit & Review a New Series



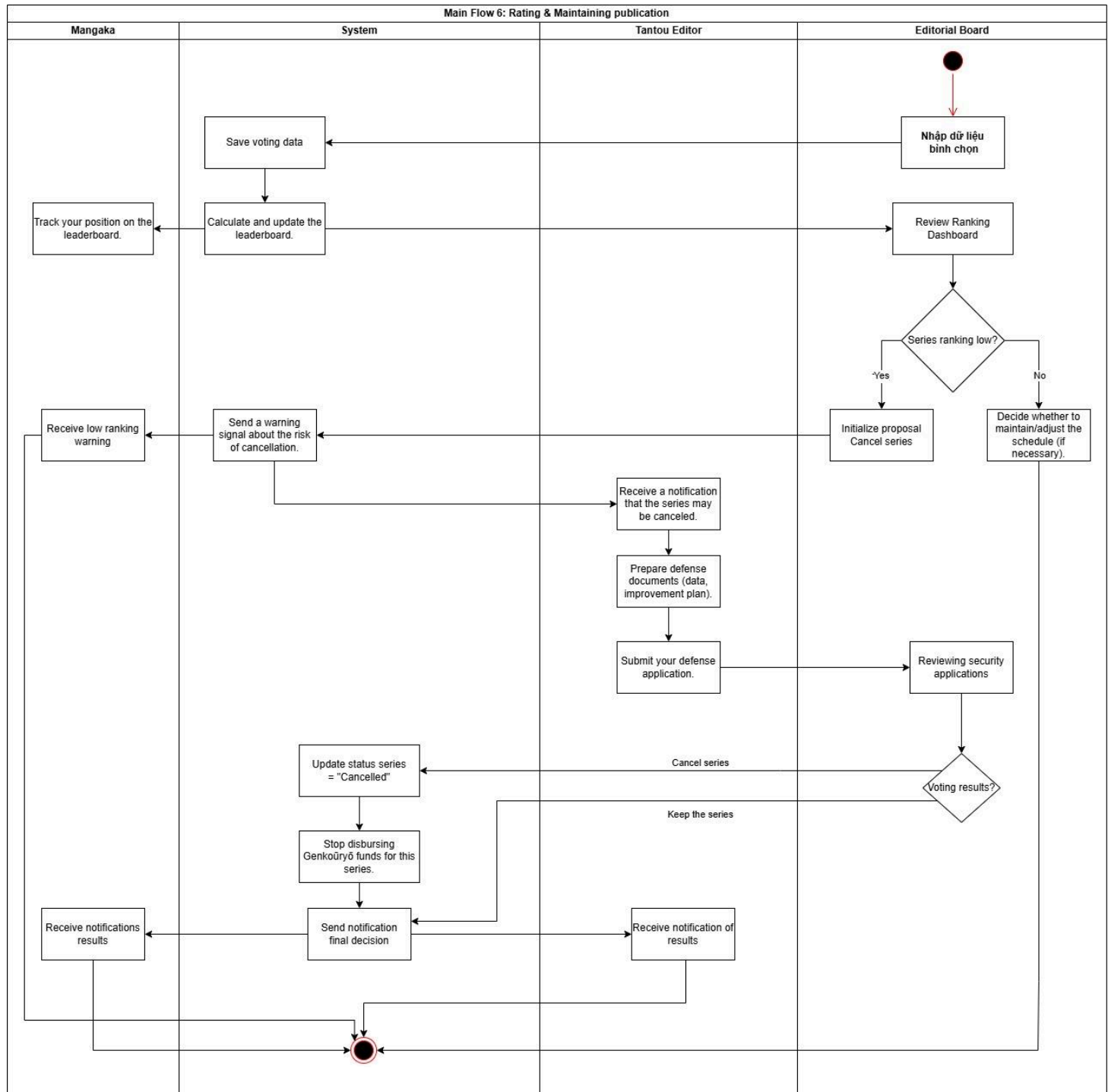
5.4 Main Flow 4: Task Assignment and Lock/Transfer Payment Processing



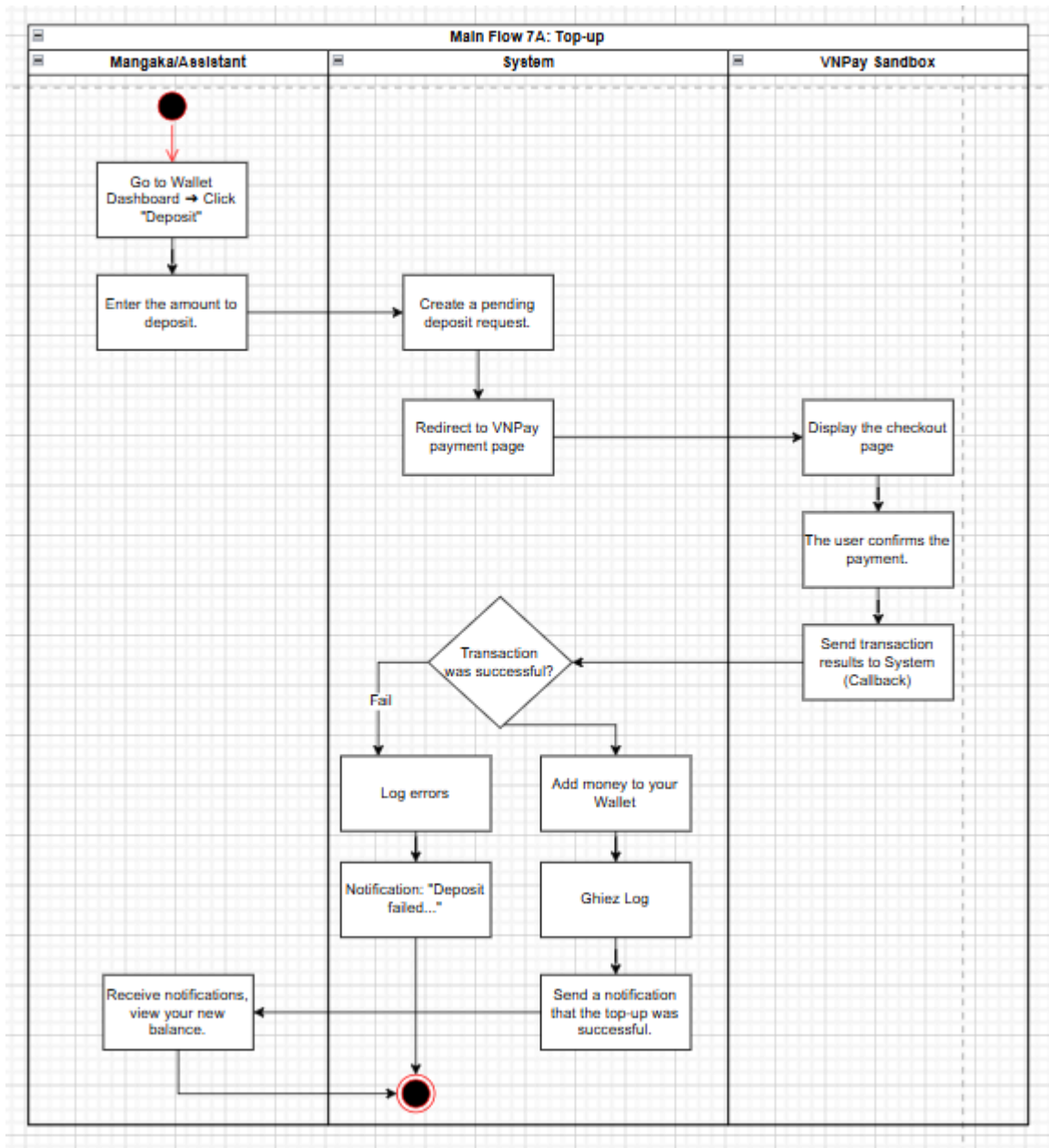
5.5 Main Flow 5: Edit Chapter & Disburse Royalties

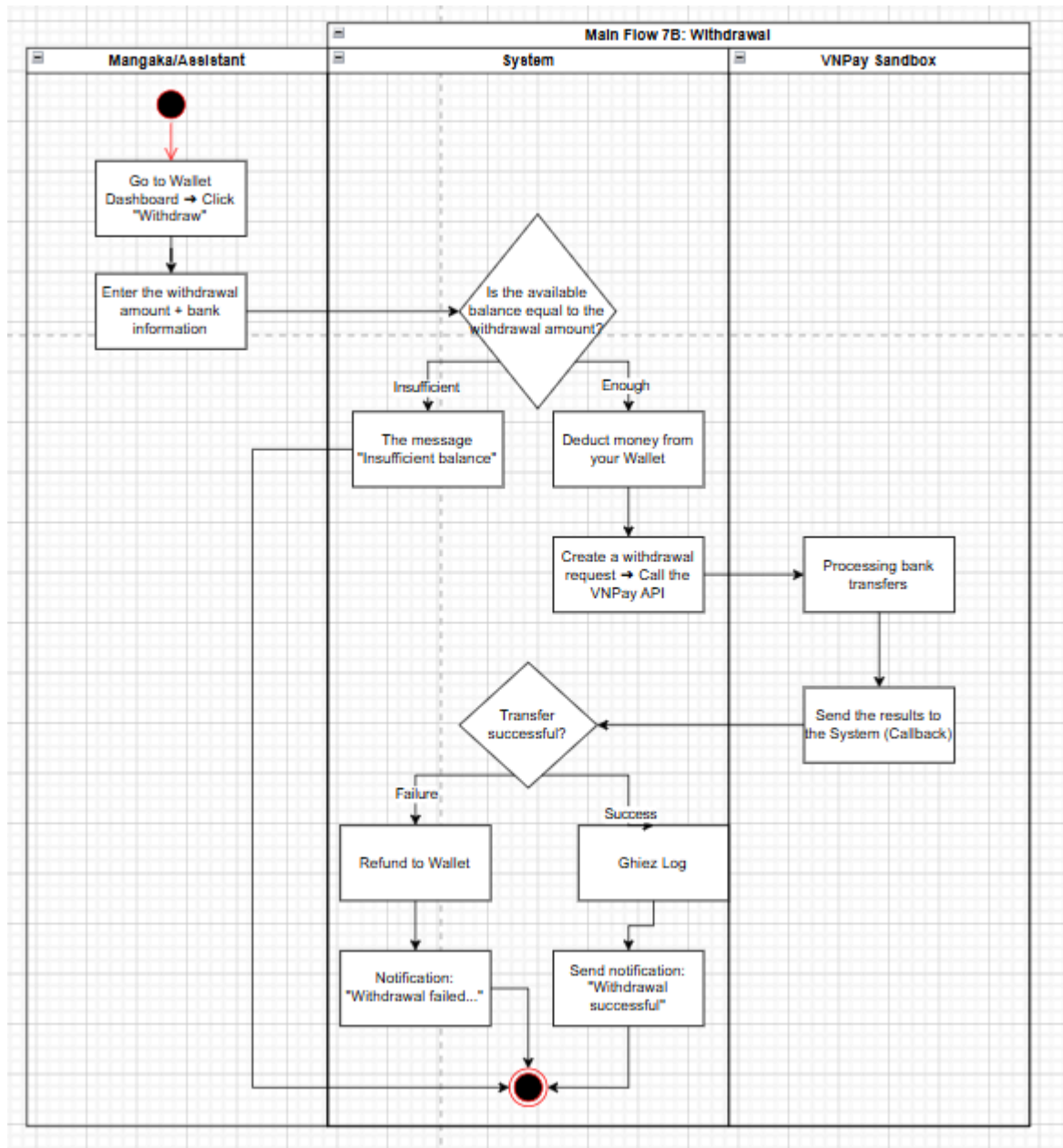


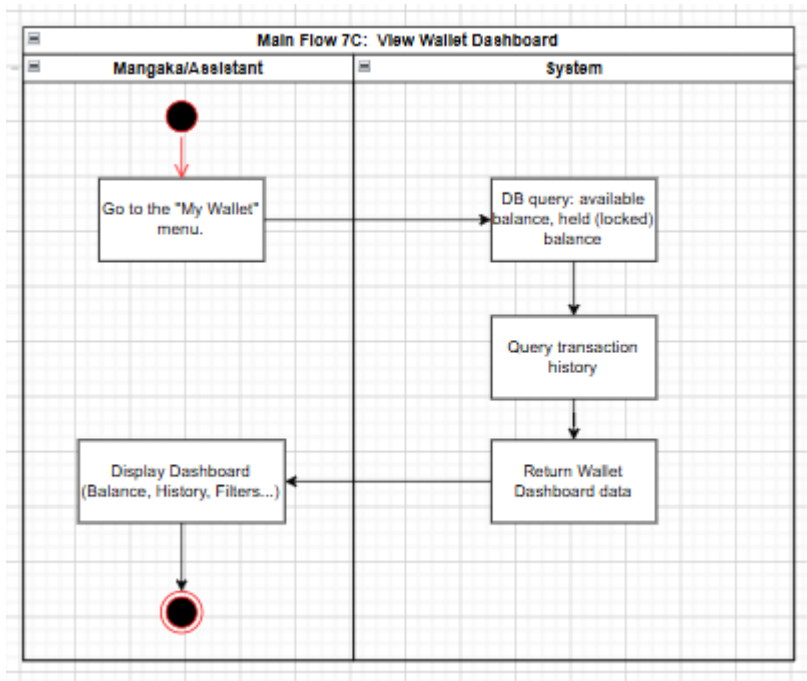
5.6 Main Flow 6: Rating & Maintaining Publication



5.7. Main Flow 7: Top-up, Withdrawal, and View Wallet Dashboard







6. Business Rules

6.1. Account & Permission Rules

ID	Description
A01	The Assistant role is the only role permitted to self-register freely from the landing page. However, the assistant account is only operational (Status = Active) after it has been explicitly reviewed and approved by a System Admin.
A02	No group, team, or workspace entities shall be designed in the database. Each author (Mangaka) directly manages their own tasks and transactions with their own Assistants (1 Mangaka - N Assistants relationship model).

6.2. Review & Funding Rules

ID	Description
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F01	Production budget for Chapter 1 is not automatically approved based on the Mangaka's proposal. The final production budget cap is decided strictly by the Editorial Board.
F02	The system only automatically credits the budget to the Mangaka's SetupFundBalance after the Editorial Board votes to Approve and the Mangaka accepts the transaction by clicking "Accept Fund".
F03	When a task is created, the system prioritizes locking funds from the SetupFundBalance. If this fund is exhausted (typically from Chapter 2 onwards), the system automatically deducts and locks the remaining amount from the WithdrawableBalance (income earned by the author). The Mangaka can only spend what they have; the system strictly prohibits overdraft or cash advances to prevent bad debts.
F04	Every real deposit or withdrawal transaction must contain a valid ReferenceCode. Periodically, the Admin imports settlement CSV files exported from VNPay for the system to automatically reconcile and flag discrepancies (mismatched Failed/Success statuses).

6.3. Task Management & Sub-Contract Rules

ID	Description
T01	A Mangaka can only create a task if their wallet balance is greater than or equal to the task fee. The fee is immediately locked in the escrow ledger and cannot be withdrawn.
T02	If the Mangaka requests a revision, the escrow funds remain locked. The Mangaka must choose an extension duration (+24 hours / 48 hours) to be added to the new task deadline.
T03a	If a task is overdue for more than 3 days without any drawing submission, the system automatically cancels the task and unlocks the escrow funds back to the Mangaka.
T03b	If an Assistant accepts a task but fails to download the template/source files within 24 hours, the Mangaka has the authority to click "Emergency Cancel" to immediately cancel the task and retrieve the escrow funds to hire another assistant.
T04	If an Assistant submits finished work and the Mangaka fails to respond within 3 days, the system automatically approves the task and transfers the funds. A warning

	notification is sent to the Mangaka on the 2nd day. Exception: If the Mangaka's profile is set to On_Leave, this countdown timer is temporarily frozen.
T05	If a series status transitions to Cancelled, the system automatically cancels all active tasks in "Pending" status and unlocks their escrow funds. For tasks currently "In-Progress", a 24-hour Grace Period is activated for Assistants to submit their work-in-progress results. After 24 hours, the Mangaka reviews the work, pays for the completed portions, and any remaining escrow balance is unlocked back to the Mangaka's wallet.
T06	When an Editor settles a task dispute, the system allows configuring a payout ratio. Based on the Editor's decision, the system automatically transfers a percentage of the escrow funds to the Assistant and unlocks the remaining percentage back to the Mangaka.
T07	Revisions submitted by Assistants must not overwrite historical files. Revisions are saved as independent TaskVersion records, allowing the Mangaka to view and approve any specific version.
T08	Assistants can submit an extension request before the task becomes overdue. If approved by the Mangaka, the new deadline is calculated as: $\text{New Deadline} = \text{Original Deadline} + \text{Requested Days}$. Each task is restricted to a maximum of 1 extension request to prevent spam.
T09	When a task transitions to its final state (Approved, Closed, or Cancelled), a background job automatically recalculates the assistant's performance metrics and updates their AssistantProfile.
T10	During task cancellation, the system must query the original lock transaction using ReferenceId, read SetupFundAmount and WithdrawableAmount fields, and automatically reverse the funds into the correct wallet allocation pockets of the Mangaka.

6.4. Review Genkōuryō Rules

ID	Description
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G01	Image compositing logic runs automatically in the background once all panel regions of a page are marked as completed.
G02	The manuscript fee (Genkōūryō) is calculated as: Genkouryo Payout = Contract Page Rate * ValidPageCount. The system calculates ValidPageCount by counting pages where IsApproved = True after the Editor's final review. However, the Editor can manually override and adjust the final page count on the QC Checklist form. The Editor must complete the QC Checklist (assessing Technical, Art, and Content criteria) to approve the manuscript. The final page count confirmed on this checklist is the single legal source of truth for payment.
G03	The system automatically disburses the calculated Genkōūryō amount into the Mangaka's wallet immediately after the Editor successfully approves the chapter.

7. Technologies

Frontend			
Technology	Version	Purpose	Relationale
React	18+	UI Framework	Component-based architecture, rich ecosystem, easy to scale.
TypeScript	5+	Type safety	Minimizes runtime bugs and improves Developer Experience (DX).
Vite	5+	Build tool	Extremely fast Hot Module Replacement (HMR) speeds, simple configuration.
React Router	6+	Routing	Built-in support for role-based routing and nested layouts.
Fabric.js	6+	Canvas manipulation	Smooth handling of drawing, region canvas selections, and annotation overlays on manga pages.
Zustand/Redux	N/A	State management	Manages complex global application states (e.g., Task Queue, Notification streams).

Frontend			
Axios	N/A	HTTP Client	Easy configuration of interceptors for JWT token attachment and Refresh Token flows.
React Query	5+	Server state	Optimizes caching strategies, supports optimistic UI updates, and implements real-time data synchronization.

Backend			
Technology	Version	Purpose	Relationale
ASP.NET Core	8+	Web API Framework	High performance execution, built-in Dependency Injection (DI) support, and native strong typing.
EF Core	8+	ORM	Simplifies database migrations, supports safe LINQ querying, and maps entity relations cleanly.
SignalR	N/A	Real-time Communication	Provides WebSocket infrastructure for live notifications and workflow progress updates.
JWT Bearer	N/A	Authentication	Stateless authentication, facilitating easy role authorization checks via claims.
AutoMapper	N/A	DTO Mapping	Ensures complete isolation between database Entities and returned API Response Models.
FluentValidation	N/A	Input Validation	Decouples validation logic from Controllers, defining clear testable validation rules.

Backend

ImageSharp	N/A	Image Processing	Supports page composition (Composite) via Alpha Blending (layering transparent graphics) based on Z-Index parameters on the server side.
VNPay SDK/API	N/A	Payment Gateway	Integrates a sandboxed environment for demonstrating secure wallet deposits and withdrawals.

Database & Storage

Technology	Purpose	Relationale
SQL Server	Primary RDBMS	Provides absolute reliability and native compatibility with the .NET ecosystem.
Redis	Cache & Session	Accelerates ranking query execution times and caches complex dashboard metrics.
Firebase Storage	Cloud File Storage	Utilizes Spark Free Tier limits to store high-resolution manga pages and resource assets.
MinIO (local)	S3-compatible storage	Emulates cloud storage containers in local development environments without cloud accounts.

AI Service

Technology	Purpose	Details
Python 3.10+	AI Programming Language	Rich Machine Learning (ML) and Deep Learning (DL) ecosystem.

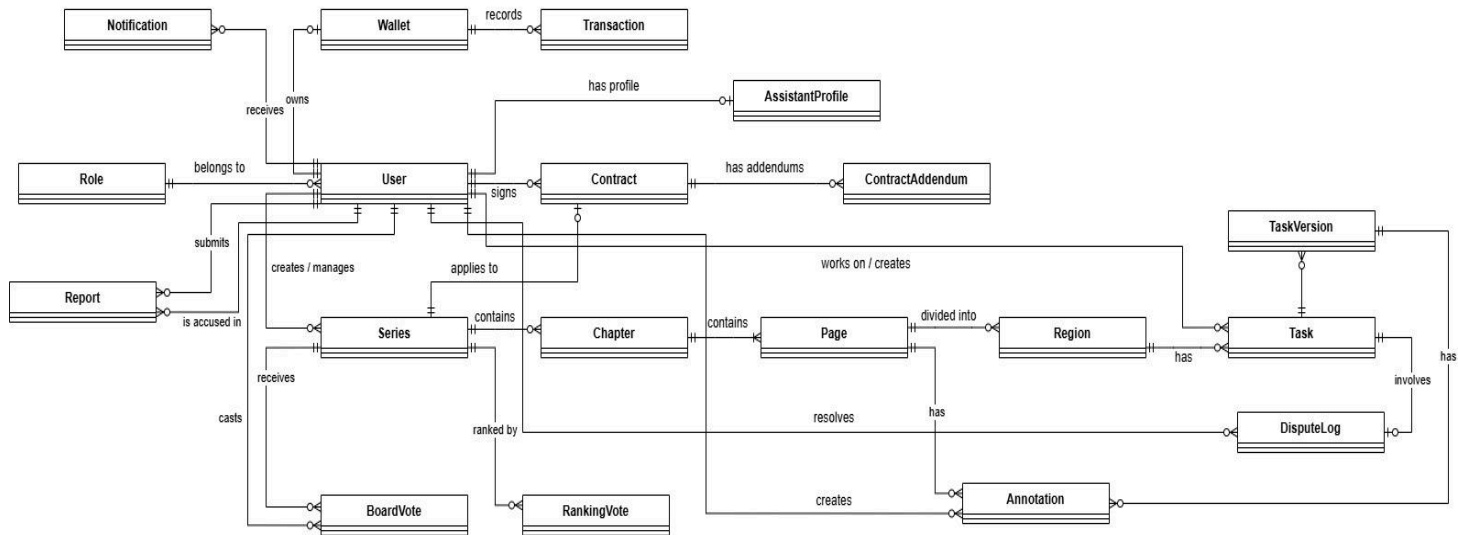
AI Service		
FastAPI	API Framework	Asynchronous request processing, auto-generated OpenAPI/Swagger documentation, high performance.
PyTorch	Deep Learning	Highly flexible framework supported by a broad AI developer community.
YOLOv8	Object Detection	Supports real-time detection of page panels (manga frames) and speech bubbles.
U-Net/SAM	Segmentation	Performs manga page layout segmentation, extracting backgrounds and character lines (Zero-shot via SAM).
OpenCV	Image processing	Performs manga page layout segmentation, extracting backgrounds and character lines (Zero-shot via SAM).

DevOps & Tooling	
Technology	Purpose
Docker/Compose	Containerization; guarantees parity between Development, Staging, and Production environments.
Github Actions	Set up automated CI/CD pipelines to build and run test cases on every code integration.
Swagger / OpenAPI	Generates visual interactive API documentation for frontend-backend integration testing.
Serilog	Structured application logging, capturing financial transaction histories for reconciliation.
xUnit / Jest	Testing frameworks used to build unit tests for the .NET Backend and React Frontend.

8. Data Model

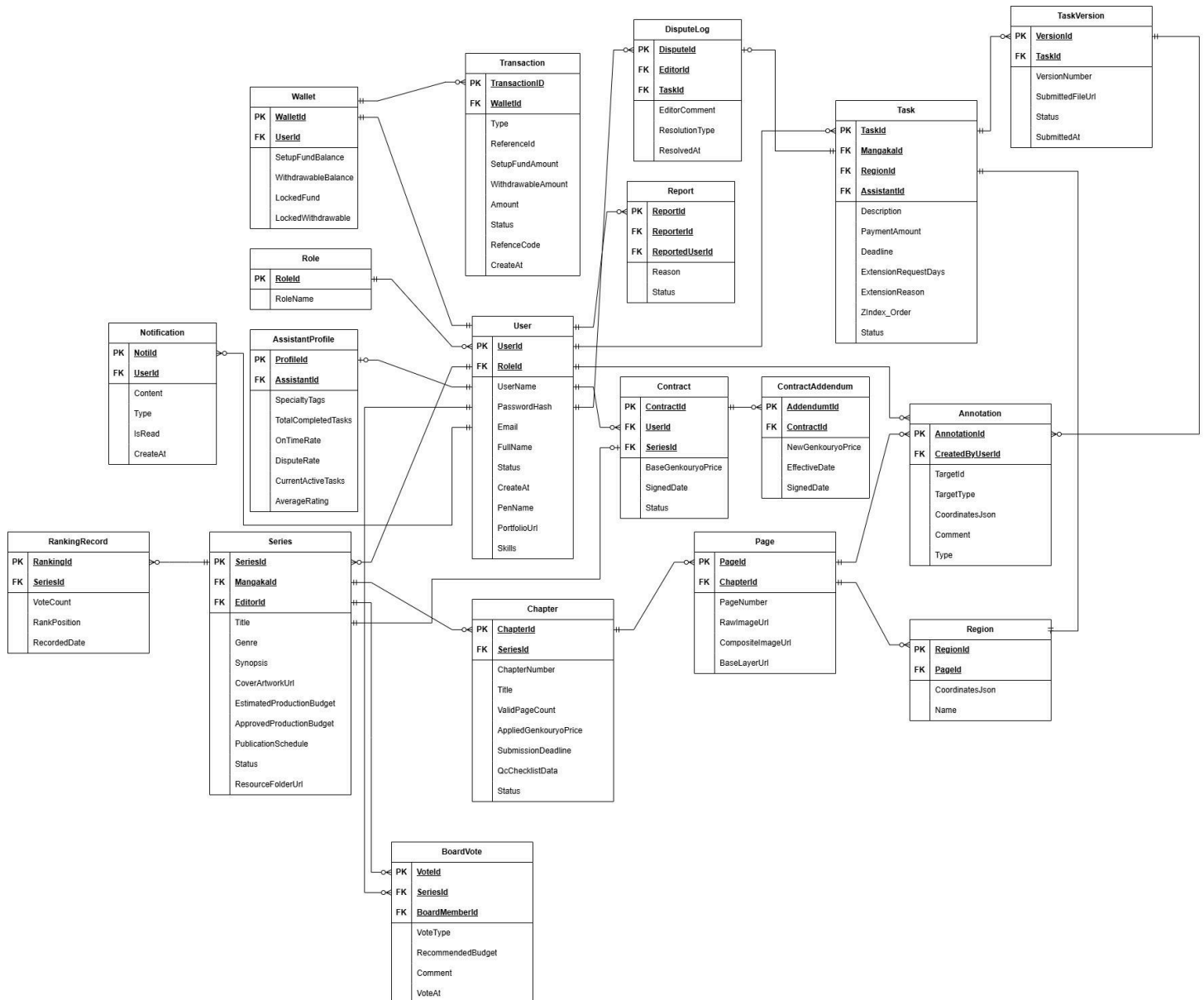
8.1. Conceptual ERD

The conceptual model displays the core entities and their high-level relationships, omitting data types and foreign key fields to focus on semantic connections.



8.2. Logical ERD

The logical model defines structural constraints, data types, primary keys (PK), and foreign keys (FK) required for database implementation.



9. User Guide

9.1. System Overview

The **Manga Creation Workflows & Publishing Management System** is a Digital Workspace platform designed for manga publishers. The system digitizes the entire workflow from creation → task assignment → review → publishing, connecting 5 main roles:

Role	Description	Access Method
System Admin	Publisher's IT Administrator.	Granted an internal account.
Tantou Editor	Editor, publisher's staff member.	Account created by Admin.
Editorial Board	Editorial board, publisher's leadership.	Account created by Admin.
Mangaka	Manga author, contracted partner.	Account created by Admin after MOU signing.
Assistant	Freelance drawing assistant.	Self-registered via the Landing Page.

9.2. System Specification

Flow 1 — Onboarding & Account Creation

For Mangaka (via off-system selection process):

- The author contacts the publisher via the Landing Page → The publisher assigns a Scout for a face-to-face meeting to evaluate their capabilities (Coachability, Speed, Teamwork, Cultural Fit, Commercial Potential).
- Both parties sign an MOU (offline document) — finalizing the unit price BaseGenkouryoPrice (VND/page).
- The Scout hands the profile over to the Admin.

Admin receives the Form → performs actions on the system:

1. Creates the Mangaka account (F5.1, F5.2).
2. Assigns a Tantou Editor in charge.

3. The Mangaka receives a login email → changes their password → begins using the system.

For Assistant (freelance registration):

1. The Assistant accesses the Landing Page → clicks "Register".
2. Fills in information: name, email, portfolio URL, and declares professional skill tags (SpecialtyTags).
3. The Admin reviews and evaluates the Portfolio → Approves the account (Status = Active) (F5.1).
4. The Assistant logs in → ready to accept tasks.

Flow 2 — Series Review & Production Funding

Mangaka logs in → Creates a series profile (F1.1):

- **Inputs:** name, genre, synopsis, and uploads sample artwork.
- **Inputs:** "Chapter 1 Production Fund" (VND) — the amount required to hire Assistants to produce the first chapter.
- Uploads "Name" (Chapter 1 storyboard/draft) (F1.2).
- Clicks "Submit for review" (F1.3) → Status: Pending/Approval.

Editor Review:

- The Editor opens the draft → uses the Annotation Tool to pin feedback (F3.1, F3.2).
- **Not Approved:** The Editor sends feedback → Mangaka revises → resubmits.
- **Approved:** The Editor compiles the profile (including the requested fund amount) → submits it to the Board (F3.3).

Board Review (F4.1):

- The Board reviews the profile + requested fund → Inputs the approved amount → clicks:
 - **Reject:** The Editor sends the reasons → Mangaka revises & adjusts the fund → Resubmits.
 - **Approve:** The system automatically changes Series.status = "Board_Approved". The Board selects the publishing schedule (Weekly/Monthly) (F4.3).

Contract Creation & Funding:

- **Admin creates Work Contract (F5.3):** Based on the signed MOU, the Admin creates a new Contract on the system (Attaching UserId, the newly approved SeriesId, and inputting the BaseGenkouryoPrice).
- **Mangaka Confirmation:** The Mangaka receives a notification, reviews the approved

Budget and Royalty Unit Price, and clicks "**Accept Fund**" → The system changes Series.status to "In Production".

- **System Allocates Funds:** The system automatically allocates funds into the Mangaka's Wallet based exactly on the amount finalized by the Board (F5.4).

Flow 3 — Production & Hiring Assistants

Mangaka assigns tasks to Assistants, with automatic payments via the Internal Wallet.

1. The Mangaka uploads manga pages (F2.1).
2. Uses the Region Selection Tool (Fabric.js) to highlight areas needing an Assistant (F2.2): background, shading, effects, etc.
3. Posts the task with a description + price + Z-index configuration (Layer stacking order to prevent overlapping when the system composites images) (F2.3) → The system checks the balance:
 - **Sufficient:** Locks the amount in the Mangaka's Wallet.
 - **Insufficient:** Mangaka deposits more money via VNPay Sandbox (F1.9).
4. The Assistant accepts the task (F2.6) → downloads the original file (F2.7) → draws → uploads the result (F2.8).
 - **[Exception - Deadline Extension]:** During the process, if unable to meet the schedule, the Assistant can submit an extension request (F2.12). If the Mangaka Approves, the system adds days to the Deadline; if Rejected, it remains unchanged (F1.10).
5. The system automatically composites the images based on Z-index coordinates once all regions have results.
6. The Mangaka reviews the composited version (F2.4) → approves, requests revisions, or cancels the task (F2.5):
 - **Approve:** The system automatically Transfers the money from the Mangaka's Wallet → Assistant's Wallet.
 - **Revision Request:** The funds remain Locked. Task.status = Revision. The Assistant revises → resubmits → Mangaka reviews again.
 - **Cancel Task:** The system Unlocks the money and refunds it to the Mangaka's Wallet. Task.status = Cancelled.

System Rule (Auto-Refund): If an Assistant has accepted a task but is 3 days past the Deadline without submitting the work (or not resubmitting a revised version), the system will automatically Cancel the task and Unlock the funds back to the Mangaka's Wallet.

System Rule (Auto-Approve): If an Assistant uploads a result and the Mangaka does not Approve or request a Revision within 3 days, the system will automatically

Approve the task and Transfer the funds to the Assistant's Wallet.

⇒ **Important Note:** When a Mangaka requests a revision, the funds are **NOT** Unlocked. If unlocked, the Mangaka could spend that money → leaving no funds to Transfer when the Assistant finishes the revision → causing a payment error. Funds are only Unlocked when a Task is completely Cancelled.

Flow 4 — Editing, Genkouryou & Publishing

The Editor inspects the chapter for errors and approves it → Mangaka receives royalties automatically.

1. The system automatically composites images into complete pages → The Mangaka reviews this auto-composited version → clicks "Submit Chapter".
2. The Editor opens the review screen and uses the Annotation Tool to pin errors on the page (F3.2). The error-catching process must be clearly categorized according to QC standards (Technical, Art, Content).
 - **Errors Found:** Editor rejects → Mangaka revises → resubmits.
 - **Approved:** Editor completes the QC Checklist to confirm the "Valid Page Count" (ValidPageCount) and clicks "Approve Chapter" (F3.6).
3. **The system automatically calculates Genkouryou (Royalties):**
 - Scans for an active Contract Addendum (ContractAddendum). If found, uses the addendum price. If not, uses the Base Price (BaseGenkouryoPrice) from the Contract.
 - Calculates: $\text{Unit Price} \times \text{Valid Page Count}$ → Automatically adds the funds to the Mangaka's withdrawable wallet.
4. The chapter is published according to the predefined schedule (Weekly/Monthly).
5. The Mangaka & Assistant can withdraw royalties/earnings from the wallet to their bank account via VNPay Sandbox (F1.8, F2.11).

Flow 5 — Ranking & Publication Maintenance

The Board monitors rankings to decide whether to maintain or cancel a series.

1. After each publishing period, reader voting data is manually entered into the system (F4.4).
2. The Ranking Dashboard updates (F4.5) → Mangaka tracks their position (F1.5).
3. **Board Evaluation:**
 - **High Ranking:** Maintain series, adjust schedule if necessary (F4.7) → Return to Flows 3 & 4.
 - **Consistently Low Ranking:** Board votes to "Cancel Series" (F4.6) → Series.status

= CANCELLED → Mangaka receives a warning (F1.6).

Rule for Series Cancellation: "When a Series transitions to the CANCELLED status, the system automatically Cancels all Tasks that are 'Pending Acceptance' or 'In Progress' and Unlocks the funds back to the Mangaka's Wallet. Simultaneously, it sends a termination notice to the relevant Assistants."

9.3. Wallet & Payment Management

The system uses an **Internal Wallet** to manage all cash flows, integrated with VNPay Sandbox for deposits and withdrawals.

Action	Actor	Description	Auto?
Production Funding	System	Allocates Chapter 1 funds to SetupFundBalance when approved by the Board.	Auto
Genkouryou Payment	System	Pays Genkouryou into WithdrawableBalance when the Editor approves the Chapter.	Auto
Task Payment	System	Locks upon posting Task → Transfers upon approval / Unlocks upon cancellation.	Auto
Deposit	Mangaka	Deposits personal funds into the wallet via VNPay Sandbox to advance hiring costs.	Manual
Withdrawal	Mangaka/ Assistant	Withdraws available funds from the wallet to a bank account via VNPay Sandbox.	Manual